

"Gross!": Personalized Video Content Moderation Using Generative Image Overlays for Aversive Videos

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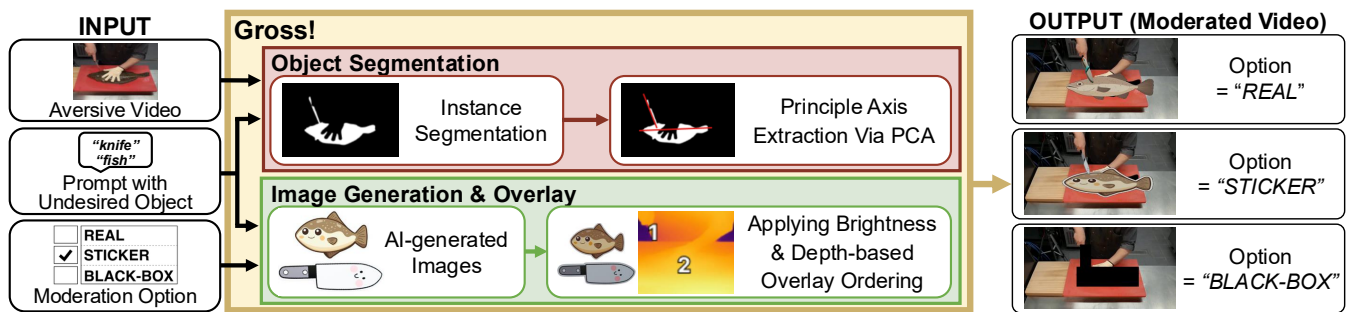


Figure 1: System overview of Gross! (Sample frame adapted from YouTube for illustrative purposes [3].)

Abstract

As the number of videos available online continues to increase rapidly, effective content filtering technologies have become essential. However, existing approaches lack personalization and struggle to preserve contextual understanding of the video. In this study, we propose a personalized content filtering method capable of understanding contextual information within videos. To evaluate the effectiveness of our approach, we conducted a user study with 18 participants. The results demonstrated an improvement in discomfort reduction and contextual understanding.

CCS Concepts

• Human-centered computing → Empirical studies in HCI.

Keywords

content moderation, personalization, contextual preserving

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1 Introduction

As digital environments continue to generate more varied video content, there is a growing need for effective filtering techniques to protect viewers from harmful or sensitive materials [1, 6, 8, 9, 16]. Despite this growing need, previous research has been limited to focusing only on how to effectively obscure undesirable objects, rather than tailoring to users' individual aversions [5, 14]. Even major video platforms apply uniform filtering policies to all users, without personalization [4, 12]. This can be particularly distressing for people with specific phobias (e.g., acrophobia, clownphobia), who may experience greater discomfort [6].

In addition, conventional filtering methods can hinder users' understanding of video context, as key visual objects are obscured or removed [7]. For example, traditional filtering methods (e.g., black box [13], pixelation [15], blurring [7, 14]) make it difficult for viewers to understand the context. To address this issue, we propose *Gross!*, a personalized filtering system that preserves contextual information for users. *Gross!* moderates aversive content by receiving user-defined aversions as text prompts and masking the corresponding objects in each video frame with AI-generated images. Through a user study with 18 participants, we observed improvements in both reducing discomfort and preserving context than the conventional method.

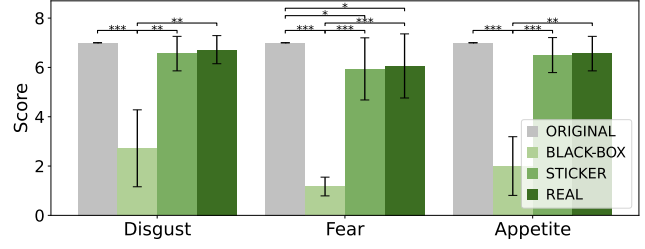
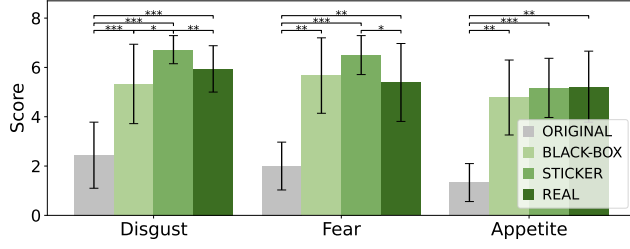


Figure 2: (Left) Content Moderation (Right) Context Preservation Effectiveness across Moderation Strategies (***) = $p < .001$; ** = $p < .01$; * = $p < .05$)

2 Implementation

2.1 Image Segmentation for Object Detection

First, we applied instance segmentation to every frame of the target video to accurately identify the location of unpleasant objects. We adopted the Grounded-SAM2 model [11], a text-prompt-based object segmentation model. Based on the user’s text input specifying undesired objects (e.g., “knife”, “fish”), the system generates precise segmentation masks for each instance. These masks define the exact regions to be filtered. Additionally, we apply Principal Component Analysis (PCA) to each mask to estimate object orientation, which guides the placement of generated replacement images.

2.2 Content Moderating Image Generation

We implemented one baseline (i.e., BLACK-BOX) and two moderation strategies (i.e., STICKER and REAL): (1) BLACK-BOX: a conventional black-box overlay for strict concealment, (2) STICKER: a playful sticker overlay designed to reduce feelings of disgust, and (3) REAL: a realistic image overlay that resembles the original object but repulsive features are removed. We used DALL-E 3 [2] to generate replacement images from users’ text input. The orientation of the generated image was calculated using PCA to align it with the orientation of the detected object in the scene. Lastly, we adjusted the brightness of the generated image to improve visual consistency with the original scene.

2.3 Image Overlay Based on Estimated Depth

To seamlessly overlay the generated images onto the scene, it is necessary to estimate depth of the images. For example, as shown in Figure 1, when both a knife and a fish are generated, the knife should be overlaid on top of the fish. We used the MiDaS [10] model, which computes relative depth from a single image. Then, the generated images were overlaid on the original image based on the detected location of the objects, calculated orientations, and estimated depth information.

3 Pilot Study

3.1 Task Design

In the experiment, we provided the following 3 scenarios: (1) Disgust-inducing (7 seconds): A video clip showing a live fish being cut with visible flapping movements intended to elicit a strong sense of disgust, (2) Fear-inducing (8 seconds): A video clip containing a clown from a horror movie, and (3) Appetite-inducing (6 seconds): A video

Version	Content Moderation Effectiveness			Context Preservation Effectiveness		
	Disgust	Fear	Appetite	Disgust	Fear	Appetite
ORIGINAL	2.44 (1.34)	2.00 (0.97)	1.33 (0.77)	7.00 (0.00)	7.00 (0.00)	7.00 (0.00)
BLACK-BOX	5.33 (1.61)	5.67 (1.53)	4.78 (1.52)	2.72 (1.56)	1.17 (0.38)	2.00 (1.19)
STICKER	6.72 (0.57)*	6.50 (0.79)	5.17 (1.20)	6.56 (0.70)**	5.94 (1.26)***	6.50 (0.71)***
REAL	5.94 (0.94)	5.39 (1.58)	5.22 (1.44)	6.72 (0.57)**	6.06 (1.30)***	6.56 (0.70)**

Table 1: Content Moderation and Context Preservation Effectiveness scores of proposed moderation strategies (STICKER, REAL) compared to BLACK-BOX (***) = $p < .001$; ** = $p < .01$; * = $p < .05$)

of a YouTuber eating fried chicken, commonly associated with stimulating appetite. Each video was processed by *Gross!* to produce three versions with different overlays (i.e., BLACK-BOX, STICKER, and REAL). For every processed videos and ORIGINAL video, we asked participants to evaluate (1) Context Moderation Effectiveness: the effectiveness of the content moderation (e.g., reduction in inducing disgust, fear, or appetite), and (2) Context Preservation Effectiveness: the degree of contextual understanding on a 7-point Likert scale after viewing each video. A higher score indicated a greater reduction in disgust, fear, or appetite, and a higher level of contextual understanding.

3.2 Procedure

We recruited a total of 18 participants (7 females and 11 males) from the local university. The average age was 28 years ($SD = 13.67$). In the experiment, we repeated the process of showing each video and conducting questionnaires evaluating the effectiveness of content moderation and context preservation. For each scenario, we provided BLACK-BOX option first to prevent exposure to the covered object by the black box. After that, the remaining versions (i.e., STICKER and REAL) were presented in a counter-balanced order to mitigate order effects. After the experiment, we conducted brief interviews with the participants.

3.3 Results

We conducted a Friedman test and pairwise Wilcoxon signed-rank tests to compare the effectiveness of context moderation and preservation between the proposed strategies (See Table 1). Among the three strategies, STICKER show the highest effectiveness for the disgust- and fear-inducing videos, while REAL outperformed others in the appetite-inducing video (See Figure 2). In terms of contextual understanding, both STICKER and REAL showed significantly higher comprehension scores than the BLACK-BOX. Notably, the two

strategies that received high ratings for moderation effectiveness were also found to preserve context relatively well.

During interviews, P5 mentioned that they preferred not to see aversive objects but still wished to understand the movie's content clearly. Interestingly, some participants pointed out that sound played a significant role in eliciting feelings of disgust, fear, or appetite. Moreover, P2, P6, and P8 noted that BLACK-BOX was perceived as even more disgusting or frightening than ORIGINAL, as it stimulated imagination. However, P2 pointed out that the high realism of REAL amplified the fear response, rather than alleviating it, under fear-inducing conditions. This finding indicates that too highly realistic sticker-style visual elements can inadvertently elicit severe aversive responses.

4 Conclusion

We propose *Gross!*, automated user-specific content moderation system that allows fine-grained filtering and preserves contextual information for the user. Our system outperformed conventional filtering methods in both content moderation and context awareness. *Gross!* demonstrates strong potential as a fully personalized filtering algorithm for video platforms.

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